Executive Summary: Six Sigma Define Phase

Order Processing Efficiency Improvement Project

Project Overview

This Six Sigma project focuses on reducing order processing cycle time and improving accuracy in our e-commerce fulfillment center. The Define Phase established the project foundation through comprehensive analysis of stakeholders, current process, customer requirements, and project scope.

Summary of Define Phase Artifacts

1. Project Charter Analysis

Overview: The Project Charter establishes formal authorization and defines project boundaries. Our charter clearly articulates the business case, problem statement, and success criteria.

Key Findings: - Current average order processing time: 4.2 days (customer requirement: 2 days) - First-pass yield: 92% (target: 99.5%) - Annual impact: 18,000 orders delayed, estimated $450,000 in lost revenue - Budget allocation: $75,000 with 4-month timeline - Estimated benefit: $1.2 million annually

Interpretation: The charter demonstrates a compelling business case with clear ROI of 16:1. Success criteria are measurable and aligned with customer expectations.

Use in Project: The charter serves as the terms of reference throughout all phases. It provides accountability and ensures scope management.

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2. Stakeholder Analysis

Overview: Identified 8 key stakeholder groups with varying influence and commitment levels.

Key Findings: - Operations Manager: High influence, enthusiastic support - Finance Director: High influence, initial skepticism (being converted to support) - Customer Service Team: Medium influence, strong support - IT Department: High influence, moderate support needed - Warehouse Staff: Direct impact, requires training support

Interpretation: Stakeholder mapping revealed IT Department as a critical success factor. Their current support level (moderate) must increase to "enthusiastic support" for implementation success.

Use in Project: Engagement plan prioritizes IT resources and addresses Finance concerns through ROI data.

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3. Stakeholder Commitment Scale

Overview: Assessed current vs. desired commitment for each stakeholder group using 8-level scale.

Key Findings: - 5 of 8 stakeholders already at desired level or above - 2 stakeholders (IT, Finance) require commitment-building activities - 1 stakeholder (Warehouse Staff) moving from “reluctant” to “willing”

Interpretation: Project readiness is strong. Focused engagement efforts on IT and Finance will mitigate risk.

Use in Project: Guides communication strategy and training emphasis.

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4. Communication Planning Worksheet

Overview: Defined communication protocols, frequency, and channels for all stakeholder groups.

Key Findings: - Weekly steering committee meetings (leadership) - Bi-weekly status reports to operations and finance - Monthly all-hands updates for warehouse staff - Daily shift briefings during implementation phase - Email escalation procedure for issues

Interpretation: Communication strategy balances information flow with operational efficiency.

Use in Project: Ensures timely escalation and stakeholder alignment throughout project.

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5. Critical-to-Satisfaction (CTS) Summary

Overview: Translated Voice of Customer into internal requirements.

Customer Requirements Identified: 1. Fast order delivery (2-day target) 2. Accurate fulfillment (no errors) 3. Proactive order status visibility 4. Simple return process 5. Reliable inventory information

Internal Translation: - CTS-1: Process cycle time 2 days - CTS-2: First-pass yield 99.5% - CTS-3: Order tracking accuracy 99% - CTS-4: Return process time 1 day - CTS-5: Inventory accuracy 99.8%

Interpretation: Five CTSs provide clear metrics for measuring project success.

Use in Project: CTSs drive Measure, Analyze, and Improve phases. They are the voice of customer embedded in project objectives.

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6. Project Plan

Overview: High-level project schedule with DMAIC phases, key activities, and milestones.

Key Activities: - **Define Phase:** Project charter, stakeholder analysis, CTS identification (Weeks 1-3) - **Measure Phase:** Current state mapping, data collection, process metrics (Weeks 4-6) - **Analyze Phase:** Root cause analysis, hypothesis testing (Weeks 7-9) - **Improve Phase:** Solution design, pilot testing, implementation (Weeks 10-14) - **Control Phase:** Process monitoring, documentation, training (Weeks 15-16)

Key Milestones: - Executive review: End of Week 3 (Define complete) - Baseline metrics approved: End of Week 6 - Root causes identified: End of Week 9 - Pilot results: End of Week 14 - Full rollout: End of Week 16

Interpretation: 4-month timeline is achievable with dedicated team. Pilot phase provides risk mitigation.

Use in Project: Provides schedule visibility and tracks progress against commitments.

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7. SIPOC Diagram

Overview: High-level process map identifying Suppliers, Inputs, Process, Outputs, and Customers.

Key Elements: - **Suppliers:** Order systems, inventory database, shipping carriers, returns processor - **Inputs:** Customer orders, inventory data, shipping requirements, return requests - **Process:** Order receipt → Validation → Picking → Packing → Shipping confirmation - **Outputs:** Shipped orders, shipping confirmations, return confirmations, performance reports - **Customers:** End customers, retailers, customer service team, management

Interpretation: SIPOC clarifies process scope and identifies key process dependencies (especially inventory data quality).

Use in Project: Provides high-level process context for Measure phase. Identifies subprocess focus areas.

Overarching Questions & Responses

a) Do you believe there was an appropriate use of tools?

Response: Yes. The tool selection was appropriate and necessary for Define Phase: - **Project Charter** provided formal authorization and clear boundaries - **Stakeholder Analysis** identified risks and engagement opportunities - **Communication Plan** prevented alignment gaps - **CTS Summary** connected customer voice to internal metrics - **SIPOC** established process context without prematurely diving into detail

The tools were used in logical sequence: establish project scope → identify stakeholders → define requirements → map process → plan execution.

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b) What tools did you use beyond each required tool? Why? What benefit?

Additional Tools Used: 1. **Voice of Customer (VOC) Interviews** (3 customer interviews, 2 manager interviews) - Why: CTS summary requires customer input beyond available data - Benefit: Identified "order visibility" as unmet need (not previously tracked) - Impact: This insight improved CTS completeness

2. **Preliminary Process Flowchart** (high-level, not detailed)
    - Why: SIPOC provided context but team needed visual process understanding
    - Benefit: Revealed 3 process handoffs causing delays
    - Impact: Identified specific Measure Phase focus areas
3. **Financial Impact Analysis**
    - Why: Project charter business case needed quantification
    - Benefit: Calculated 16:1 ROI, secured executive support
    - Impact: Finance moved from skepticism to support

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c) Do you believe the project is ready to move to the next phase? Why or why not?

Response: YES, the project is ready to move to Measure Phase with the following evidence:

Readiness Indicators: 1. Executive sponsor approval obtained 2. Project charter signed by all stakeholders 3. Team fully assigned (1 Black Belt, 2 Green Belts, 3 process experts) 4. Budget approved ($75,000) 5. Clear, measurable objectives established 6. Customer requirements documented (CTS) 7. High-level process understood (SIPOC) 8. Communication plan in place 9. Stakeholder commitment at desired levels for 6 of 8 groups 10. Critical dependencies identified (IT, inventory data)

Confidence Level: 85% confident in project success based on Define Phase artifacts and stakeholder alignment.

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d) If the project is not ready, what measures need to be taken to recover?

Not applicable - project is ready. However, contingency measures if issues emerge:

1. **If IT commitment decreases:**
    - Escalate to CIO; offer IT recognition/resources
    - Target: 1-week resolution
    - Impact: Minimal if addressed quickly
2. **If customer requirements shift:**
    - Formal change control process triggered
    - CTS summary updated
    - Target: 2-week assessment, potential 1-2 week delay
    - Benefit: Ensures project remains customer-focused
3. **If team resources become unavailable:**
    - Identify and train backup team members during Measure Phase
    - Target: 2-week cross-training
    - Impact: Prevents implementation delay

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e) Does the project charter, problem, scope, or other aspects need refinement? Please explain.

Response: Minor refinements recommended, but no major revisions needed:

Charter Refinement Needed: 1. **Clarify "inventory accuracy" scope** (currently includes supplier data; should we focus only on internal warehouse?) - Action: Add one sentence clarifying scope boundary - Timeline: Pre-Measure Phase - Impact: Prevents scope creep

2. **Define success criteria more precisely**
    - Current: "Process cycle time 2 days"
    - Recommended: "95% of orders processed within 2 days; 100% within 3 days"
    - Reason: Accounts for outliers and customer tolerance
    - Impact: More realistic success definition
3. **Add risk section** (currently not included)
    - Key risks: IT system changes, holiday season volume spikes, staff turnover
    - Mitigation: Communication plan, pilot phase, training documentation
    - Timeline: Document in first week of Measure Phase

Scope Refinements: - Current scope includes returns processing (may be better as separate project) - Recommendation: Keep in scope for Define Phase analysis, decide in Analyze Phase if sub-problem - Rationale: Returns likely interconnected with order fulfillment delays

Overall Assessment: Project definition is solid. Recommended refinements are enhancements, not corrections.

Conclusion

The Define Phase has successfully established project foundation. The project demonstrates: - Clear business justification (16:1 ROI) - Strong stakeholder alignment (75% at desired commitment) - Customer-focused objectives (CTS defined) - Realistic timeline and budget - Identified success metrics

Project is approved to proceed to Measure Phase.

Document Date: December 2025 **Project Manager:** Aditya Kalambe